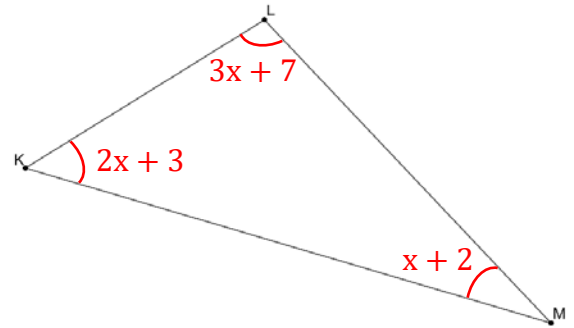


Use this information to answer questions 1 through 5.

Triangle KLM is shown, where

$m\angle K = (2x + 3)^\circ$, $m\angle L = (3x + 7)^\circ$, and $m\angle M = (x + 2)^\circ$.



1. Find the value of x .

$$(2x + 3) + (3x + 7) + (x + 2) = 180$$

$$6x = 168$$

$$x = 28$$

2. Identify the measure of $m\angle K$.

$$m\angle K = 2(28) + 3 = 59^\circ$$

3. Identify the measure of $m\angle L$.

$$m\angle L = 3(28) + 7 = 91^\circ$$

4. Identify the measure of $m\angle M$.

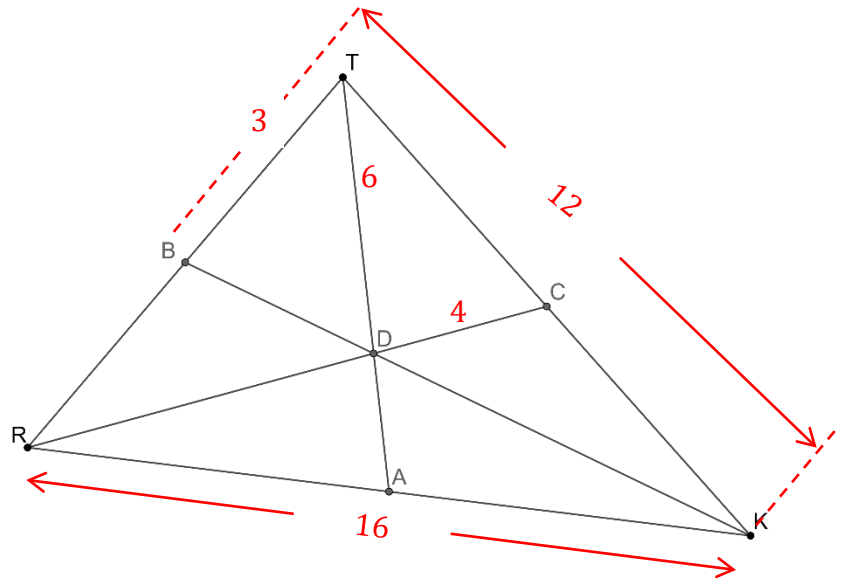
$$m\angle M = 28 + 2 = 30^\circ$$

5. List the sides of the triangle in order of their lengths, smallest to largest.

$\overline{KL}$, $\overline{LM}$, $\overline{KM}$

Use this information to answer questions 6 through 10.

Triangle RTK is shown, where $\overline{RC}$, $\overline{BK}$, and $\overline{AT}$ are median. $BT = 3$, $BK = 12$, $DC = 4$, $TD = 6$, and $RK = 16$.



6. Find the length of $\overline{BD}$.

$$BD = \frac{1}{3}(BK) = \frac{1}{3}(12) = 4$$

7. Find the length of $\overline{RC}$.

$$RC = 3(DC) = 3(4) = 12$$

8. Find the length of $\overline{DA}$.

$$DA = \frac{1}{2}(TD) = \frac{1}{2}(6) = 3$$

9. Find the length of $\overline{RT}$.

$$RT = 2(BT) = 2(3) = 6$$

10. Find the perimeter of triangle RBD .

$$\begin{aligned} P &= RB + BD + RD \\ &= 3 + 4 + 8 = 15 \end{aligned}$$